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ME 3902
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Test 2 Task A Report

In this experiment, I utilized a schematic and code from ADArC (figure 1) [1] as a basis to build a system that triggers a servo (Smraza S51 Micro Servo) when an ultrasonic sensor (HC-SR04) detects something within a distance range. When the sensor records a distance between 10cm and 16cm, it rotates the servo from 0° to 180° and back (90° baseline).

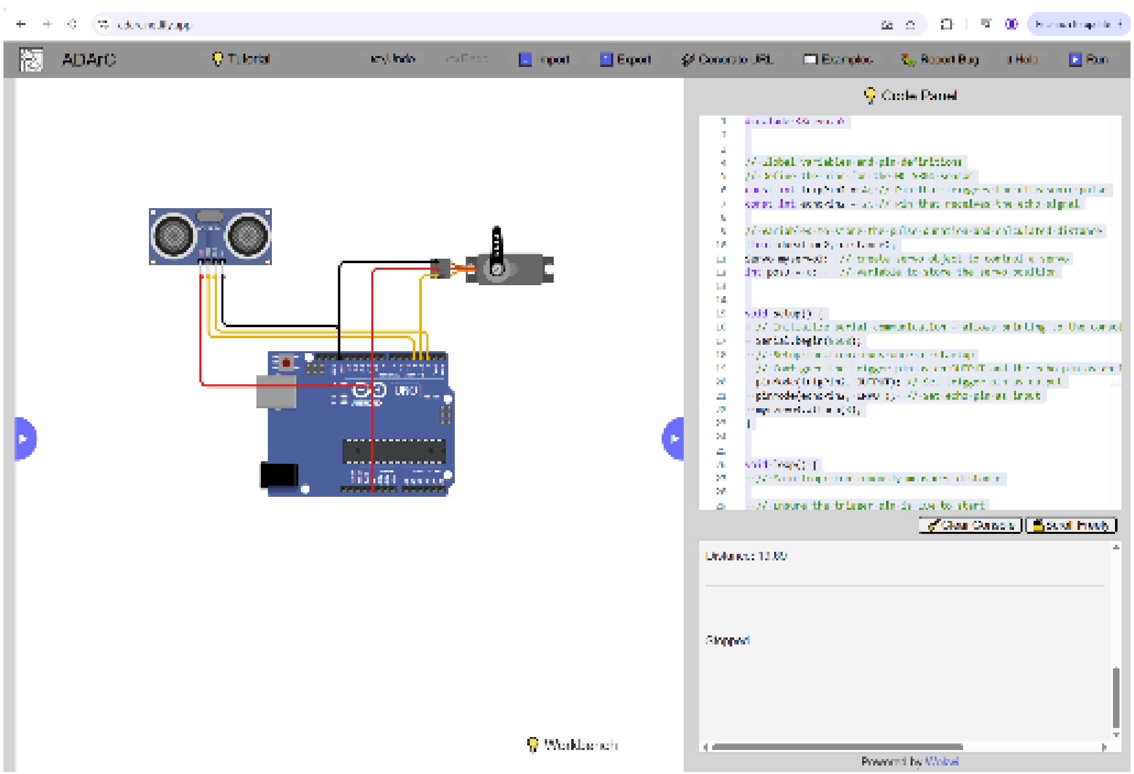


Figure 1

This preliminary design was then slightly modified to incorporate an OLED screen (SSD1306) that could display the measured distance observed by the ultrasonic sensor. This new circuit was designed in Wokwi [2] such that it could be put together on a full-size breadboard (figure 2). The original ADArC code was inserted into Wokwi before being modified to accommodate the OLED panel. The panel updates in near real-time (100ms delay) to the distance recorded by the ultrasonic sensor, to enable simple and quick data collection for the experiment. The experiment consisted of comparing the ultrasonic sensor's recorded distance from the servo to the actual distance as measured by a tape measure laid out on a flat surface in front of the sensor (figure 3). Data points were then gathered and compiled in a table to determine how the ultrasonic sensor's "measured" values differed from the concrete measurements along the tape measure. This table (figure 4) was then compared against the output after the Arduino was restarted, allowing for an evaluation of the circuit's repeatability, of which it was found to be consistent.

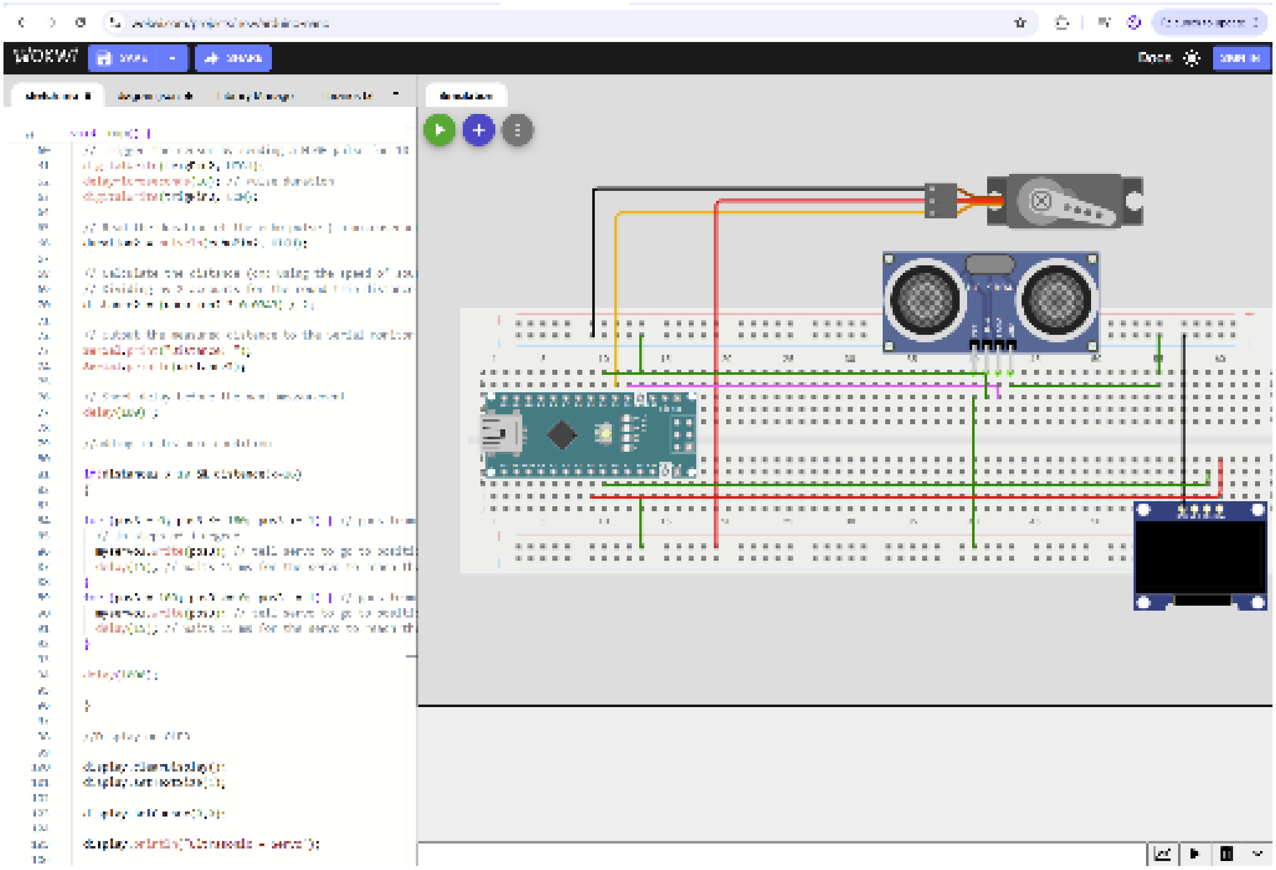
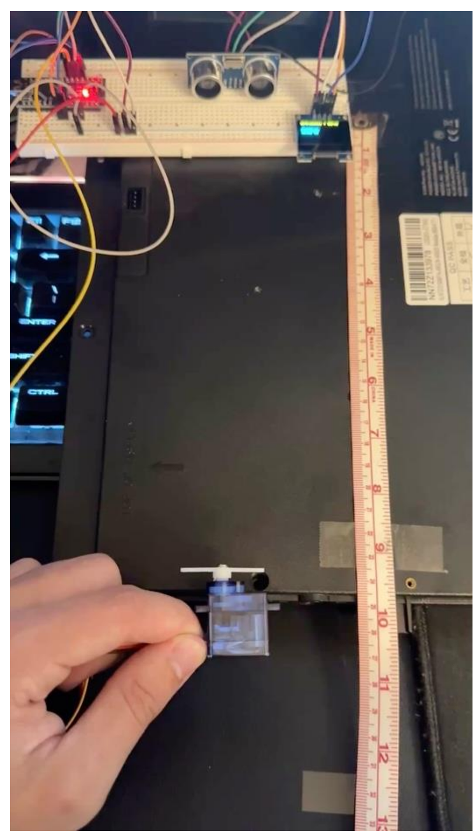


Figure 2 (above), Figure 3 (right) & Figure 4 (below)



Trial #	Actual value (cm)	Measured value (cm)	Difference (cm)
1	9.5	9.16	0.34
2	3.5	3.23	0.27
3	4.5	5.06	-0.56
4	6.25	6.53	-0.28
5	11	12.36	-1.36

In conclusion, while there was some inaccuracy visible as shown by the table when it came to the sensor versus a control measurement tool like a measuring tape, resetting the system and measuring again did not produce error greater than what was already occurring. The main fix I would consider is increasing the polling rate of the sensor and dropping the screen's delay so that it is more responsive and doesn't take as long to "settle in" to an accurate measurement.

Github Link:

https://github.com/CharlieJSmith57/ME-3902-E1-2026/blob/main/Test_2/Testing_Ultrasonic_Servo_Arduino

Works Cited

[1] "ADArC," [Online]. Available: adarc.netfly.app. [Accessed 9 06 2026].

[2] "Wokwi," [Online]. Available: wokwi.com. [Accessed 8 6 2026].

[3] rkprad, "ME 3902 - Testing Input/Output devices using Arduino Nano - Test 2 UltraSonic, Servo and OLED," 4 6 2026. [Online]. Available: https://www.youtube.com/watch?v=jbxLbfi90Ao&t=2059s. [Accessed 8 6 2026].